ENTHUSE, LEADER & ACHIEVER COURSE

PHASE : ALL SRG ENTHUSE, LEADER, ACHIEVER COURSE & LAKSHYA PHASE

TARGET : PRE-MEDICAL 2026

Test Type : **SRG-MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 21-05-2026

ANSWER KEY

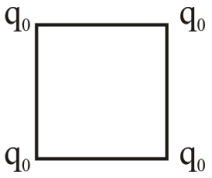
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Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	4	4	2	3	3	3	1	4	3	1	2	2	2	1	4	2	3	2	2	3	1	4	3	3	2	2	3	3	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	4	4	1	2	3	2	1	1	3	4	1	4	2	2	2	4	1	4	3	1	1	4	2	3	4	3	4	1	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	2	4	1	2	2	4	4	2	1	3	3	4	2	3	4	3	1	2	3	2	3	2	1	1	2	3	3	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	1	3	4	2	3	4	2	3	2	1	1	2	3	3	4	3	2	2	2	2	2	2	1	2	2	2	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
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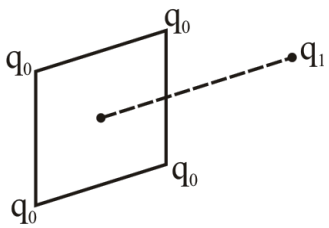
HINT – SHEET

1. Ans (3)

Solution :

q_1 is in unstable equilibrium if q_0 & q_1 have same signs.





As charge moves toward the (+) ve z-axis or –ve z-axis, then net force on charge will be along the displace direction& would not come ever.

q_1 is in stable equilibrium if q_1 & q_0 have difference signes

In this case as charge moves away from centre then net force on the charge opposite to displace direction.

2. Ans (3)

Resultant field near surface of conductor is $\frac{\sigma}{\epsilon_0}\hat{n}$.

3. Ans (2)

$$V_{surface} = \frac{Kq}{R} = V_B$$

4. Ans (1)

Work done by electric field :

$$W = q\,[V_i - V_f] = q$$

$$\left[\frac{KP\cos 45°}{(1\times 10^{-2})^2} - \frac{KP\cos 135°}{(2\times 10^{-2})^2}\right]$$

$$\text{work done} = \frac{100}{3}\ J$$

5. Ans (2)

Electric field inside cavity

$$\vec{E} = \frac{\rho\vec{c_1c_2}}{3\varepsilon_0}$$ which is uniform throught the cavity

6. **Ans (4)**

Total energy of system

$= kE + \text{electrostatic PE} + \text{spring pE}$

$= \left(\frac{1}{2}mV^2 + \frac{1}{2}mV^2\right) + \frac{kQQ}{r} + \frac{1}{2}k(r-\ell_0)^2$

$= mV^2 + \frac{Q^2}{4\pi\epsilon_0 r} + \frac{1}{2}k(r-\ell_0)^2$

7. **Ans (2)**

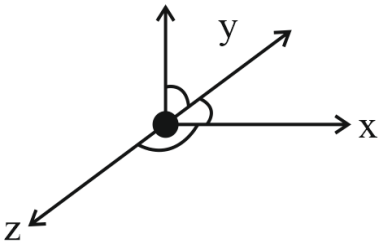
$\vec{E} = -\left[\frac{\partial v}{\partial x}\hat{i} + \frac{\partial v}{\partial y}\hat{j} + \frac{\partial v}{\partial z}\hat{k}\right]$

$= -\left[(12-6xy)\hat{i} + \left(-3x^2+2z^2\right)\hat{j} + (4y)\hat{k}\right]$

$\left(\vec{E}\right)_{(1,0,-2)} = -\left[12\hat{i} + 5\hat{j}\right]$

$E = \sqrt{144+25} = \sqrt{169} = 13 \text{ unit}$

8. **Ans (2)**



$\phi = \vec{E}.\left(100\times10^{-4}\right)\left(\cos\alpha\hat{i} + \cos\beta\hat{j} + \cos\gamma\hat{k}\right)$

$\phi = \left(3\times10^3\hat{i}\right).\left(100\times10^{-4}\right)\left(\frac{1}{2}\hat{i} + \frac{1}{2}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}\right)$

$\phi = 3\times10^3\times100\times10^{-4}\times\frac{1}{2}$

$\phi = 15\frac{Nm^2}{C}$

9. **Ans (3)**

The resistance between points BD and CD are 2Ω.

$R_{ABD} = R_{ACD} = 4+2 = 6\Omega$

$R_{eq} = \frac{6\times6}{6+6} = 3\Omega$

10. **Ans (2)**

$v_d = \frac{i}{p\cdot s\cdot q}$

11. **Ans (2)**

$R = \frac{\rho\ell}{A} \Rightarrow R\times\frac{1}{A}\times\frac{1}{r^2}$

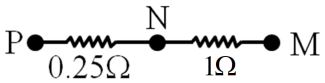
$\frac{R_{AB}}{R_{CA}} = \left(\frac{r_{CA}}{r_{AB}}\right)^2 = \left(\frac{2a}{a}\right)^2 = 4$

$\frac{V_{AB}}{V_{CA}} = \frac{R_{AB}}{R_{CA}} = 4$

12. **Ans (2)**

Given from data

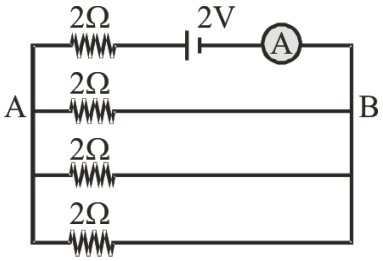
$V_P - V_M = 4\times1 = \text{Volt}$



$V_N - V_M = \frac{1}{1.25}\times4 = 3.2\text{volt}$

13. **Ans (2)**

The circuit can be redrawn as follows.



Resistance across AB

$R_{AB} = \frac{2}{1+1+1} = \frac{2}{3}\ \Omega$

Total resistance of circuit

$R_T = 2 + \frac{2}{3} = \frac{8}{3}\ \Omega$

Current through ammeter

$i = \frac{2}{(8/3)} = \frac{6}{8} = \frac{3}{4}A$

14. **Ans (1)**

$i = \frac{E_{net}}{R_{net}} = \frac{6+6}{3+2+R} = \frac{12}{5+R}$

$(V)_A = E_A - ir_A = 0$

$E_A = ir_A$

$\Rightarrow 12 = 10 + 2R$

$R = 1\Omega$

15. **Ans (4)**

For balanced wheatstone bridge the resistance connected between B and D are meaningless hence R_2 have any finite value for balancing condition.

$\frac{10}{R_1} = \frac{30}{9} \Rightarrow R_1 = 3\Omega$

16. **Ans (4)**

As power source is connected

so V = const

$E = \frac{V}{d} \Rightarrow E = \frac{300}{5\times10^{-2}} = 6000\frac{V}{m}$

17. Ans (2)

Force of attraction between plates,

$$F = \frac{Q^2}{2A\varepsilon_0} = \frac{1}{2}\varepsilon_0 AE^2$$

where $E = \frac{V}{d}$

$$F = \frac{\varepsilon_0 AV^2}{2d^2}$$

So $F^1 = \frac{K\varepsilon_0 AV^2}{2d^2}$ & $F^1 = KF$

18. Ans (2)

$$C = \frac{A\varepsilon_0}{d - t + \frac{t}{K}}$$

given $t = \frac{d}{2}$, $K = 3$

$$C = \frac{3A\varepsilon_0}{2d}$$

19. Ans (2)

Formula : $K' = \frac{U_{big}}{n.u_{small}}$

Calculation : $\frac{n^{5/3}u}{n.u} = n^{2/3}$

20. Ans (1)

Stress = Y strain

$$\frac{F}{A} = Y\,\alpha\Delta Q$$

$$F = YA\,\alpha\,\Delta Q$$

21. Ans (2)

$$\frac{m_A C_A}{m_B C_B} = \frac{(4/3)\pi r_A^3 \rho_A C_A}{(4/3)\pi r_B^3 \rho_B C_B}$$

$$= \left(\frac{r_A}{r_B}\right)^3 \frac{\rho_A C_A}{\rho_B C_B}$$

$$= \left(\frac{1}{2}\right)^3 \times \left(\frac{2}{1}\right) \times \left(\frac{1}{3}\right) = \frac{1}{12}$$

22. Ans (4)

ΔQ required $= MS\Delta\theta$

$$= 3000g \times 1\ \frac{cal}{g°c} \times (77 - 27) \times 4.2$$

$$= 6.3 \times 10^5 J$$

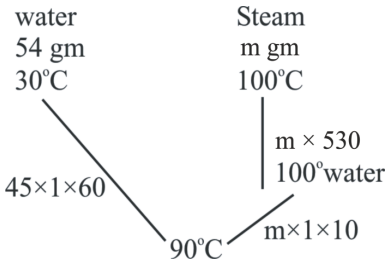
$\because$ 4×10^4J heat combustion by fuel = 1g

$\therefore$ 1 J heat combustion by fuel $= \frac{1}{4 \times 10^4}$g

$\therefore$ 6.3×10^5 J heat combustion by fuel

$$= \frac{1}{4 \times 10^4} \times 6.3 \times 10^5 = 15.75 \text{ g/min.} \approx 16 \text{ g/min}$$

23. Ans (2)



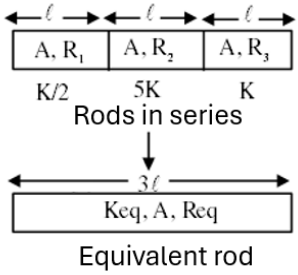
$m \times 530 + m \times 10 = 54 \times 1 \times 60$

$540\ m = 54 \times 60$

$$m = \frac{54 \times 60}{540} = 6$$

$M_{mix} = 54 + 6 = 60$ gm

24. Ans (1)



For series combination $\longrightarrow$

$R_{eq} = R_1 + R_2 + R_3$

$$\frac{3\ell}{AK_{eq}} = \frac{\ell}{\frac{KA}{2}} + \frac{\ell}{5KA} + \frac{\ell}{KA}$$

$$\frac{3}{K_{eq}} = \frac{2}{K} + \frac{1}{5K} + \frac{1}{K}$$

$$\frac{3}{K_{eq}} = \frac{10 + 1 + 5}{5K}$$

$$\frac{3}{K_{eq}} = \frac{16}{5K}$$

$$K_{eq} = \frac{15}{16}K$$

25. Ans (4)

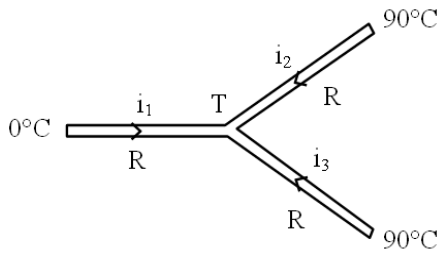
According to Kirchhoff's law

$i_1 + i_2 + i_3 = 0$

$$\frac{0 - T}{R} + \frac{90 - T}{R} + \frac{90 - T}{R} = 0$$

$3T = 180$

$T = 60°C$



26. Ans (4)

$P \propto T^4 - T_0^4$

$\frac{P_2}{P_1} = \frac{T_2^4 - T_0^4}{T_1^4 - T_0^4}$

$\frac{P_2}{60} = \frac{(1200)^4 - (400)^4}{(800)^4 - (400)^4} = \frac{16}{3}$

$P_2 = 320$ W

27. Ans (3)

$PV = nRT$

$P = \frac{nRT}{V}$

For A :-

$\Delta P = \frac{m_A RT}{MV} - \frac{m_A RT}{2V}$

$\Delta P = \frac{-m_A RT}{2MV}$ ....(1)

For B :-

$1.5\Delta P = \frac{-m_B RT}{2MV}$ ....(2)

(1) ÷ (2)

$\frac{1}{1.5} = \frac{m_A}{m_B}$

$\frac{2}{3} = \frac{m_A}{m_B}$

$3m_A = 2m_B$

28. Ans (3)

For $\gamma = \frac{7}{5}$ $C_\upsilon = \frac{R}{\gamma - 1} = \frac{R}{\frac{7}{5} - 1} = \frac{5R}{2}$

$C_p = \frac{\gamma R}{\gamma - 1} = \frac{(7/5)R}{\frac{7}{5} - 1} = \frac{7R}{2}$

For $\gamma = \frac{4}{3}$ $C_\upsilon = 3R$, $C_p = 4R$

$\therefore \quad \gamma_{mix} = \frac{\frac{7}{2} + 4}{\frac{5}{2} + 3} = \frac{15}{11}$

29. Ans (3)

adiabatic process $\left[P_1^{\frac{1-\gamma}{\gamma}} T_1 = P_2^{\frac{1-\gamma}{\gamma}} T_2\right]$

$(2)^{\frac{1-7/5}{7/5}} \times 300 = (1)^{\frac{1-7/5}{7/5}} \times T_2$

$(2)^{-\frac{2}{7}} \times 300 = T_2$

$T_2 = \frac{300}{2^{2/7}} = 244$ K

$T_2 = 244 - 273 = -29°C$

30. Ans (2)

$VP^3 =$ constant

we know that $P = \frac{nRT}{V} \Rightarrow P \times \frac{T}{V}$

so $V \times \frac{T^3}{V^3} = k$

$\frac{T_1^3}{V_1^2} = \frac{T_2^3}{V_2^2}$

$[T_1 = T, V_1 = V, V_2 = 27V, T_2 = ?]$

$T_2^3 = \left(\frac{27V}{V}\right)^2 \times T_1^3$

$T_2^3 = 27 \times 27 \times T_1^3$

$T_2 = 3 \times 3 \times T = 9T$

31. Ans (2)

$\frac{(\Delta Q)_p}{(\Delta Q)_v} = \frac{C_p}{C_v}$

$\Rightarrow \frac{310}{(\Delta Q)_v} = 1 + \frac{2}{3}$

$\Rightarrow \frac{310}{(\Delta Q)_v} = \frac{5}{3}$

$\Rightarrow 310 \times \frac{3}{5} = (\Delta Q)_v$

$\Rightarrow 62 \times 3 = (\Delta Q)_v$

$\Rightarrow \Delta Q_v = 186$ Joule

32. Ans (4)

SHM speed relation : $v^2 = \omega^2 (A^2 - x^2)$.

At mean position $v_{max} = \omega A = 16$ cm/s. With A = 4 cm,

$\omega = \frac{16}{4} = 4s^{-1}$, $\omega^2 = 16$.

Given $v = 8\sqrt{3}$cm/s $\Rightarrow v^2 = 192$.

use $v^2 = \omega^2 (A^2 - x^2)$:

$192 = 16(4^2 - x^2) = 16(16 - x^2) \Rightarrow 12 = 16 - x^2$

$x^2 = 4 \Rightarrow x = 2$ cm.

33. Ans (4)

$T = 8$ sec, $\omega = \frac{2\pi}{T} = \pi/4$, A = 1 cm

$x = A \sin \omega t \Rightarrow x = 1 \sin\left(\frac{\pi}{4} \times \frac{4}{3}\right)$.

$x = \sin \pi/3 \Rightarrow \left[x = \frac{\sqrt{3}}{2}\right]$ at $t = \frac{4}{3}$ sec.

$a = -\omega^2 x$

$a = -\frac{\pi^2}{16} \times \frac{\sqrt{3}}{2} \Rightarrow a = -\frac{\sqrt{3}}{32}\pi^2$

34. Ans (2)

Given : $U = U_0(1 - \cos \alpha x)$

$F = -\frac{dU}{dx}$

$\therefore \quad F = -\frac{d}{dx}[U_0(1 - \cos\alpha x)] = -U_0\alpha\sin\alpha x$

As αx is small, so $\sin\alpha x \sim \alpha x$

$\therefore \quad F = -U_0\alpha^2 x \qquad .....(i)$

$F \propto x$ and $-$ve shows that F is directed towards the mean position, hence the body executes SHM.

For SHM, $F = -kx \qquad .....(ii)$

Comparing (i) and (ii), we get

$k = U_0\alpha^2$

Time period of oscillation,

$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{m}{U_0\alpha^2}}$

35. Ans (3)

Potential Energy in SHM: $U = \frac{1}{2}kx^2$

Kinetic Energy in SHM: $K = \frac{1}{2}k\left(A^2 - x^2\right)$

Total Energy in SHM: $E = \frac{1}{2}kA^2 = 20J$

i. $U_{x=0}$: $U = \frac{1}{2}k(0)^2 = 0J$, so i matches with r (0 J)

ii. $K_{x=0}$: Since total energy is 20 J and U = 0J, K = E $-$ U = 20 $-$ 0 = 20J, so ii matches with p (20J)

iii. $U_{x=A/\sqrt{2}}U = \frac{1}{2}k\left(\frac{A}{\sqrt{2}}\right)^2 = \frac{1}{2}k.\frac{A^2}{2} = \frac{E}{2} = \frac{20}{2} = 10J$, so iii matches with q (10J)

iv. $U_{x=A/2}$: $U = \frac{1}{2}k\left(\frac{A}{\sqrt{2}}\right)^2 = \frac{1}{2}k.\frac{A^2}{2} = \frac{E}{4} = \frac{20}{4} = 5J$, so iv matches with s (5J)

The correct matches are:

i-r (0 J)

ii-p (20 J)

iii - q (10 J)

iv-s (5 J)

36. Ans (3)

$v = \sqrt{\frac{T}{m}}$, but $v = n\lambda \quad \therefore \quad \lambda = \frac{1}{n}\sqrt{\frac{T}{m}}$

As the transverse wave travels upwards along the rope, there is neither any change in frequency nor in mass per unit length m.

$\therefore \quad \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_1}{T_2}}$

At the lower end, $T_1 = 2\times9.8$ N, $\lambda_1 = 0.06$m

At the top end, $T_2 = 8 \times 9.8$ N

$\therefore \quad \frac{0.06}{\lambda_2} = \sqrt{\frac{2\times9.8}{8\times9.8}} = \frac{1}{2}$

$\therefore \quad \lambda_2 = 0.12$ m

37. Ans (3)

Amplitude of reflected wave:

$\frac{A_r}{A_i} = \sqrt{\frac{E_r}{E_i}}$

$A_r = 10 \times \sqrt{0.81} = 10 \times 0.9 = 9$

Reflection from a denser medium introduces a phase change of π.

Mathematically, this changes the sign of the amplitude: $A_r \longrightarrow -A_r$.

The incident wave (ax + bt) travels in the negative x-direction.

The reflected wave must travel in the positive x-direction, changing (ax + bt) to (ax $-$ bt).

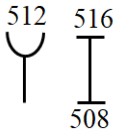
38. Ans (1)

Laplace corrected newton's formula by assuming that sound propagation in air is an adiabatic process, not isothermal. This means there is no exchange of compression and rarefaction. For a process to be adiabatic, heat must not have time or means to flow. This is true because.

$\longrightarrow$ Air is poor conductor of heat

$\longrightarrow$ Sound wave travel very fast.

39. Ans (4)

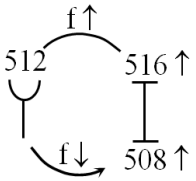


beat frequency

4

Since with increase in Tension frequency of piano string increases

So, after T↑



2

since decreasing condition is satisfied by 508

so Ans 508 Hz

40. Ans (3)

$$1\left(\frac{V}{4L_C}\right) = 3\left(\frac{V}{2L_O}\right)$$

$$\frac{1}{2L_C} = \frac{3}{L_O}$$

$$L_O = 6\,L_C$$

$$L_O = 6\ (20\ cm)$$

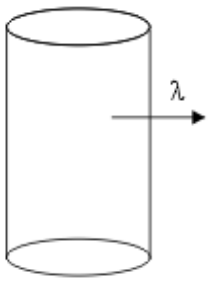
$$L_O = 120\ cm$$

41. Ans (1)

$$E_A = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$E_B = 0$$

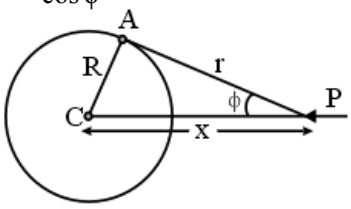
$$qE_{net} = \frac{mV^2}{r} \Rightarrow \frac{q\lambda}{2\pi\epsilon_0 r} = \frac{mV^2}{r} \Rightarrow \sqrt{\frac{q\lambda}{2\pi\epsilon_0 m}}$$



42. Ans (2)

$$\cos\phi = \frac{r}{x}$$

$$x = \frac{r}{\cos\phi}$$



∵ conducter is equipotential

$$\therefore V_A = V_C = \frac{KP}{x^2}$$

$$= \frac{KP}{\left(\frac{r}{\cos\phi}\right)^2} = \frac{KP\cos^2\phi}{r^2}$$

43. Ans (2)

Question Explanation:

Calculating the equivalent resistance of an infinite ladder network.

Concept:

In an infinite ladder, adding or removing one repeating unit does not change the total resistance (R_{eq}).

Solution:

Let the total equivalent resistance between A and B be X.

Because the network is infinite, the resistance of the entire circuit to the right of the first vertical branch is also X.

We can redraw the circuit as the first three resistors (top horizontal R, bottom horizontal R, and vertical R) connected to an equivalent resistor X.

The vertical R and the rest of the ladder X are in parallel. Their equivalent is:

$$R_p = \frac{RX}{R+X}$$

This combination is in series with the two horizontal R resistors. So:

Let $R_{AB} = x\Omega$

$$R_{AB} = R + \frac{Rx}{R+x} + R$$

$$x = 2R + \frac{Rx}{R+x}$$

$$\Rightarrow x = 2(1+\sqrt{3})k\Omega$$

Final Answer:

Option (2)

44. Ans (2)

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

45. Ans (1)

Explanation of question :- The question asks for equivalent resistances between corners 1–2 and 1–3 respectively in the given tetrahedron network of wires.

Concept : Combination of Resistors

Solution :

The figure will look like when seen from above

each edge has resistance = r

equivalent resistance

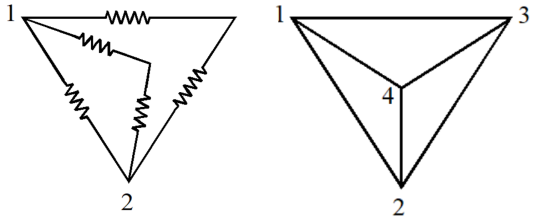
= between 1 and 2

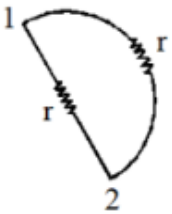
(resistance between 3 and 4

will not work as the given

configuration behaves

like Wheatstone bridge)





$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{r} + \frac{1}{r} \Rightarrow \frac{2}{r} \text{ or } R_{eq} = \frac{r}{2}.$$

equivalent resistance between 1 and 2

equivalent resistance between 1 and 3

can be find out same process

$$R_{eq} = \frac{r}{2}.$$

Option A is Correct.

Final Answer :- Option (1)

46. Ans (4)

$r = K[A]^{1/2}$

as reaction are of fractional order reactions then reactions must be a complex reactions.

47. Ans (2)

$r = K[NO]^a [Cl_2]^b$

$$3 \times 10^{-3} = \left(\frac{0.15}{0.05}\right)^b \Rightarrow b = 1$$

$$9 \times 10^{-3} = \left(\frac{0.15}{0.05}\right)^a \Rightarrow a = 2$$

48. Ans (3)

Question Explanation: Find rate law for reaction.

Concept: This question is based on mechanism of reaction.

Solution:

If step (2) is slowest step $r = K_2 [N_2O_2] [H_2]$

$$\frac{K_f}{K_b} = \frac{[N_2O_2]}{[NO]^2}$$

$$[N_2O_2] = \frac{K_f}{K_b}[NO]^2$$

$$r = K_2 \times \left(\frac{K_F}{K_b}\right)[H_2][NO]^2$$

$$r = K [H_2] [NO]^2$$

49. Ans (2)

Larger the value of Ea, Reaction will more sensitive towards temperature.

50. Ans (2)

Question Explanation : Calculate concentration of M^{+3}.

Concept : Nernst Equation.

Solution :

$$Pt \mid \underset{0.01M}{M^{+2}}(aq), \underset{xM}{M^{+3}}(aq) \mid\mid \underset{0.04M}{Cu^{+2}}(aq) \mid Cu(s)$$

$$E^\circ_{cell} = 0.34 + 0.77 = 1.11 \text{ V}$$

$$E_{cell} = E^\circ_{cell} - \frac{0.06}{2}\log\frac{[M^{+3}]^2}{[M^{+2}]^2} \times \frac{1}{[Cu^{+2}]}$$

$$1.17 = 1.11 - 0.03 \log \frac{[x]^2}{[0.01]^2} \times \frac{1}{[0.04]}$$

$$0.06 = -0.03 \log \frac{[x]^2}{[0.01]^2} \times \frac{1}{[0.04]}$$

$$-2 = \log \frac{[x]^2}{[0.01]^2} \times \frac{1}{[0.04]}$$

$$10^{-2} = \frac{x^2}{4 \times 10^{-6}}$$

$$x = 2 \times 10^{-4} M$$

51. Ans (3)

$Q = I \times t = 2 \times 60 \times 60$

$w = \frac{63.5 \times 2 \times 60 \times 60}{2 \times 96500} = 2.37g$

$\%\ \text{efficiency} = \frac{0.237}{2.37} \times 100 = 10\%$

or

$W = \frac{Eit}{96500}$

$0.237 = \frac{63.5}{2} \times i \times \frac{3600}{96500}$

$(i)_{used} = 0.2A$

$\text{Current efficiency} = \frac{(i)_{used}}{(i)_{passed}} \times 100$

$= \frac{0.2}{2} \times 100 = 10\%$

52. Ans (1)

$Na_2SO_4 \rightleftharpoons 2Na^+ + SO_4^{2-}$

$H_2O \rightleftharpoons H^+ + OH^-$

H^+ , OH^- are discharged at respective electrodes.

So, pH remains same i.e.= 7

So colourless

Hence ans is A

53. Ans (4)

Oxidizing agent $\propto$ SRP

54. Ans (3)

EMF of cell in an intensive property.

55. Ans (3)

Question Explanation:

Question is asking about the equivalent conductance of the electrolyte at infinite dilution

Concept:

This question is based on application of Kohlrausch law

Solution:

$K_a = C\alpha^2$

$\sqrt{\frac{K_a}{C}} = \alpha$

$\sqrt{\frac{25 \times 10^{-6}}{0.01}} = 5 \times 10^{-2}$

$\alpha = 0.05$

$\alpha = \frac{\Lambda}{\Lambda^\circ}$

$\Lambda^\circ = \frac{19.6}{0.05} = 392$

Final Answer:

The correct option is (3)

56. Ans (2)

$ClO_3^- + 2H_2O + 4e^- \rightarrow ClO^- + 4\bar{O}H\ ;\ \Delta G_1^\circ$

$ClO^- + H_2O + e^- \rightarrow \frac{1}{2}Cl_2 + 2\bar{O}H\ ;\ \Delta G_2^\circ$

$\frac{1}{2}Cl_2 + e^- + Cl^-\ ;\ \Delta G_3^\circ$

$ClO_3^- + 3H_2O + 6e^- \rightarrow Cl^- + 6\bar{O}H\ ;\ \Delta G^\circ$

$\Delta G^\circ = \Delta G_1^\circ + \Delta G_2^\circ + \Delta G_3^\circ$

$-6 \times F \times E^\circ = -4 \times F \times 0.54 - 1 \times F \times 0.45 - 1 \times F \times 1.07$

$E^\circ = \frac{3.68}{6} = 0.61\ V$

Option : (2)

57. Ans (2)

When $p_A^o > p_B^o$

then $y_A > x_A$

58. Ans (3)

$P_T = P_A^0 + \left(P_B^0 - P_A^0\right) X_B \Rightarrow P_A^0 = 100$,

$= P_B^0 = 260$ units

59. Ans (3)

$[\text{Glucose}] = \frac{15 \times 1.2 \times 10}{180} = 1M$

60. Ans (1)

Urea & $A\ell_2(SO_4)_3$ are non-volatile solute :

$\frac{P_A^0 - P}{P_A} = x_B.i \Rightarrow P = (1 - x_B\ .\ i).\ P^\circ_A$

$i \uparrow \Rightarrow$ V.P.(P) $\downarrow$

"i" for urea = 1 & for $Al_2(SO_4)_3 = 5$

V.P. in container - 1 > V.P. in container - 2

After opening stopper gas (vapours) will travel from high pressure to low pressure region.

Thus, vapours from container-1 enter into container-2 after opening stopper.

$\Rightarrow$ Weight of container-2 containing $Al_2(SO_4)_3$ increases.

61. Ans (3)

Correct Ans. is (3)

62. Ans (4)

$V_{H_2O} = 1L$

$\Rightarrow W_{H_2O} = 1kg \quad [\because d_{H_2O} = 1\ g/mL]$

$\Rightarrow \Delta T_f = m.kf$

$\Rightarrow \dfrac{\Delta T_f}{k_f} = m = \dfrac{1}{1000}$

$\Rightarrow m = \dfrac{n_{glucose}}{W_{H_2O(kg)}} = \dfrac{n}{1} = \dfrac{1}{1000}$

$\Rightarrow n = \dfrac{1}{1000}$ [$\because$ M_w of glucose = 180 g/mol]

$\Rightarrow W_{glucose} = nM_w = \dfrac{1}{1000} \times 180 = 0.18\ g$

$\Rightarrow W_{glucose} = 0.18\ g$

$\Rightarrow$ option [D]

63. Ans (4)

$$\underset{\text{L.R.}}{Ba(OH)_2} + 2HCN \longrightarrow Ba(CN)_2 + 2H_2O$$

20 millimol 50 millimol $\Delta H = -24kJ/mol$

heat evolved will be $= 24 \times 20 \times 10^{-3}$

$= 48 \times 10^{-2}$ kJ

480 Joule

64. Ans (1)

Question Explanation: Calculating the enthalpy change for the hydration of an anhydrous salt using dissolution data.

Concept: Enthalpy of Hydration (ΔH_{hyd})

Solution:

We are given,

$BaCl_2.2H_2O(s) + aq \longrightarrow BaCl_2(aq) \longrightarrow \Delta_{sol}H_1 = 8.8$

$BaCl_2(s) + aq \longrightarrow BaCl_2(aq) \longrightarrow \Delta_{sol}H_2 = -20.6$

But, we have to write below equation for hydration of $BaCl_2$.

$BaCl_2(s) + 2H_2O \longrightarrow BaCl_2.2H_2O(s)$

$\Delta_{hyd}H^\circ = ?$

So, $\Delta_{hydration}H = \Delta H_2 - \Delta H_1$

$= -20.6 - 8.8$

$= -29.4$ kJ/mol

Final Answer: Option (1)

65. Ans (2)

Question Explanation: Explain the reason for the high stability of lyophilic colloids.

Concept: Stability of Lyophilic Sols

Solution: **Lyophilic colloids** (solvent-loving) are stable because of the strong interaction between the dispersed phase and the dispersion medium.

This interaction leads to the formation of a dense, stable layer of the **dispersion medium (solvation layer)** around the colloidal particles, which protects them from aggregation and coagulation.

Final Answer: A layer of dispersion medium on their particles

66. Ans (3)

Question Explanation: Predict the order of coagulating power of anions for a positive sol.

Concept: Hardy-Schulze Rule

Solution:

$Fe(OH)_3 \longrightarrow$ Positive sol and it is coagulated by Anion.

Coagulating power $\propto$ charge on anion

Order $\longrightarrow PO_4^{3-} > SO_4^{2-} > NO_3^-$

Final Answer: The correct option is (3).

67. Ans (2)

Explain the question:

Determine activation energy from two rate constants at different temperatures.

Concept:

Two-Point Arrhenius Equation

Solution:

$H_2 + I_2 \rightarrow 2HI$ $\ell n\dfrac{K_2}{K_1} = \dfrac{E_A}{R}\left[\dfrac{1}{T_1} - \dfrac{1}{T_2}\right]$

$\ell n\dfrac{1}{2.5 \times 10^{-4}} = \dfrac{E_A}{R}\left[\dfrac{1}{600} - \dfrac{1}{800}\right]$

$\log_{10}\left(\dfrac{1.0}{2.5 \times 10^{-4}}\right) = \dfrac{E_a}{2.303 \times 8.314}\left(\dfrac{800-600}{600 \times 800}\right)$

$3.602 = \dfrac{E_a}{19.147} \times \left(\dfrac{200}{480000}\right)$

Solving for *Ea*: $Ea \approx 165696$ J mol^{-1}. Converting to kJ mol^{-1} yields 165.696 kJ mol^{-1}.

Final Answer:

The correct option is (2)

68. **Ans (1)**

$A \longrightarrow B + C$

$P_0 \quad 0 \quad 0$

$P_0 - x \quad x \quad x$

$0 \quad p_0 \quad p_0$

Given :

$p_0 - x = 200$

$p_0 + x = 300$

$\Rightarrow p_0 = 250$

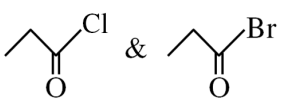
$x = 50$

$K = \frac{1}{t} \ell n \frac{p_0}{p_0 - x}$

$= \frac{1}{600} \ell n \frac{250}{250 - 50} \sec^{-1}$

$= \frac{1}{600} \ell n\, 1.25 \sec^{-1}$

69. **Ans (1)**

 & are not isomer

Mixed question

70. **Ans (3)**

Geometrical isomerism

71. **Ans (4)**

Question Explanation :

The task is to assign R/S configurations to three Fischer
projections of 1-chloro-1-phenylethane and determine their stereochemical relationship.

Concept :

Cahn-Ingold-Prelog priority, Chiral center, Fischer projection, Enantiomers,
Identical compounds

Solution :

Priority: Cl(1) > Ph(2) > CH_3(3) > H(4).

. Compound (I): 1 ⟶ 2 ⟶ 3 is clockwise. H is horizontal, so configuration is S.

. Compound (II): 1 ⟶ 2 ⟶ 3 is counter-clockwise. H is horizontal, so configuration is R.

. Compound (III): 1 ⟶2 ⟶ 3 is clockwise. H is horizontal, so configuration is S.

Comparing configurations: (I) and (II) are opposite (S vs R), making them enantiomers. (I) and
(III) are the same (S vs S), making them identical.

Final Answer : Option (4)

72. **Ans (1)**

Incorrect protonation

73. **Ans (4)**

Question Explanation: Ordering the basic character of substituted anilines.

Concept: Basicity

Solution: The basicity of an aniline derivative is determined by the availability of the lone pair on the nitrogen atom. The methyl group (CH3) is an electron-donating group, which increases basicity. The nitro group (NO2) is an electron-withdrawing group, which decreases basicity. The correct order is IV > III > II > I. **Calculation**: N/A

Final Answer: 3

74. **Ans (2)**

p-nitrophenol is more acidic than o-nitrophenol

75. **Ans (2)**

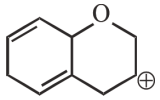
Question Explanation :

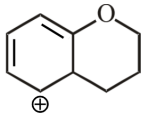
Question is asking about most stable cation.

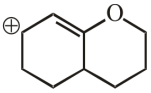
Concept :

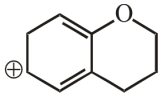
This question is based on

Solution :

(1)  $\alpha - C - H = 4$ (Incomplete octat)

(2)  Complete octate (more stable) ($6\pi e^-$ reso)

(3)  Complete octate (more stable) ($4\pi e^-$ reso)

(4)  Incomplete octate $4\pi e^-$

Resonance $\alpha - C - H = 2$

Stability carbon cation : (2) > (3) > (4) > (1)

Final Answer:

Option (2)

76. Ans (2)

Question Explanation:

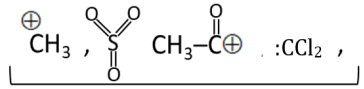
The question asks to count the number of electrophiles among the given species .

Concept:

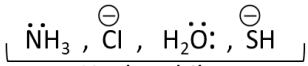
Attacking reagent and Electrophile.

Solution:

- Electrophiles are e^- deficient species .
- Generally $\oplus$ve charged , vacant orbital and non polar λ –bonded species are art as electrophiles .



Electrophiles .

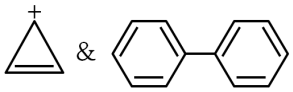


Nucleophile .

So, there are (4) Electrophiles are present in given spears .

Final Answer: option (2)

77. Ans (4)

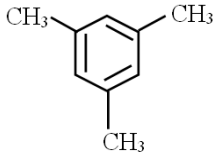


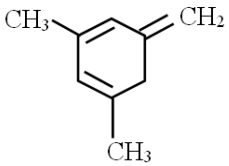
are aromatic

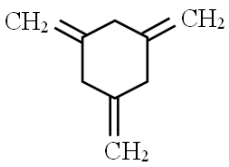
78. Ans (1)

Solution :- Heat of Hydrogenation [HOH] $\propto$

$$\text{Resonance} \propto \frac{1}{\text{stability of Alkene}}$$

(1)  → most stable due to Aromaticity

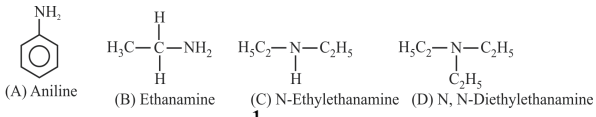
(2)  → stable due to resonance

(3)  → Least stable due to absence of resonance

→ ∴ Order of stability of Alkene :- 1 > 2 > 3

Answer :- Order of HOH :- 3> 2 > 1 (III > II > I)

79. Ans (4)



(A) Aniline (B) Ethanamine (C) N-Ethylethanamine (D) N, N-Diethylethanamine

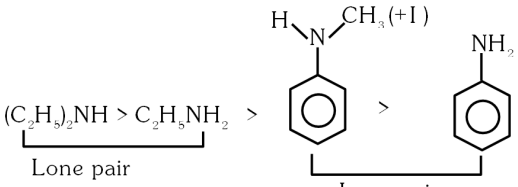
$$\text{Basic Strength} \propto \frac{1}{pK_b}$$

Order for pK_b : A > B > D > C

80. Ans (3)

Basic strength order

N-ethylethanamine > Ethanamine > N-methylaniline > Benzenamine

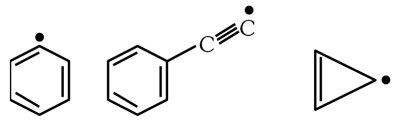


Lone pair localised

Lone pair delocalised

81. Ans (1)

$$\text{Bond dissociation energy} \propto \frac{1}{\text{Stability of free radical formed after C - H bond fission homolytically}}$$



I I III

Stability order III > I > II

Bond dissociation energy order ⇒ II > I > III.

82. Ans (1)

$F_2 > Cl_2 > Br_2 > I_2$

decreasing order of reactivity

83. Ans (4)

Question Explanation - Identify the specific chemical reagent sequence that fails to transform isobutane into the alkene isobutylene.

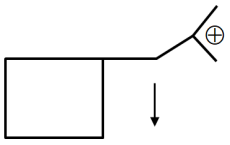
Concept - Hydroboration and Elimination reactions.

Solution -

Option 4 involves hydroboration-oxidation; however, this process requires an initial alkene, not an alkane like isobutane, to produce the alcohol necessary for dehydration. Therefore, this sequence cannot start with isobutane.

Final Answer - Option (4)

84. Ans (2)



No Rearrangement

85. Ans (3)

Explaining the Question :- Question ask for Matching the items in List-I with the appropriate purification methods in List-II

Concept:- Purification methods

Solution :- (A) Coloured impurity: (R) Charcoal treatment :Activated charcoal adsorbs coloured impurities (decolorization)

(B) o-Nitrophenol + p-Nitrophenol: (P) Steam distillation o-Nitrophenol is steam-volatile (due to intramolecular H-bonding), while p-nitrophenol is not.

(C) Crude Naphtha: (Q) Fractional distillation Naphtha is a mixture of hydrocarbons separated by boiling point differences.

(D) Glycerol + Sugars: (S) Distillation under reduced pressure Glycerol is heat-sensitive; vacuum distillation lowers its boiling point.

Final Answer :- Option 3

86. Ans (4)

Explanation Question:

The question asks which compound gives a positive Lassaigne's test for nitrogen?

Concept:

This question is based on Detection of elements-Lassaigne's test.

Solution:

• Lassaigne's test detact nitrogen by converting it into NaCN. which gives a prussian blue colour with $FeSO_4$.

• To form sodium cyanide, both carbon and nitrogen must be present in the compound.

• Ethylamine ($CH_3-CH_2-NH_2$) contains in a form that gives NaCN – Positive test.

• Hydrazine (H_2N-NH_2)-Lack of carbon-Negative test.

• Nitro compound and Pyridine does not give Lassagine's test

So. Ethylamine given positive lassaigne's test.

Final Answer: (4)

87. Ans (3)

Question Explanation:

Match the given processes and tests in Column-I with their correct descriptions in Column-II.

Concept:

Chromatography, Lassaigne's test for N and S, Steam distillation, Vacuum distillation.

Solution:

Column I	Column II
A. Best and latest technique for isolation, purification and separation of organic compounds	(i) Chromatography
B. In Lassaigne's test for 'N'	(iv) Prussian blue coloured complex is formed
C. In Lassaigne's test for 'S'	(iii) Violet coloured complex is formed
D. Aniline is purified from water by	(v) Steam distillation
E. Glycerol is purified from spent lye	(ii) Vacuum distillation

The correct match is A-(i) B-(iv) C-(iii) D-(v) E-(ii).

Final Answer: option (3)

88. Ans (4)

Rate of Nitration $\propto$ +M , +H, +I group

89. **Ans (1)**

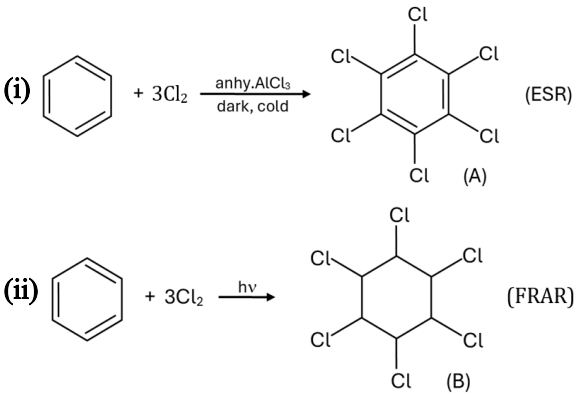
Question Explanation :

Identify the products of two distinct reaction involving benzene and chlorine gas under different reaction conditions?

Concept :

This question is based on Electrophilic substitution reaction (ESR), Free radical addition reaction. (FRAR)

Solution :



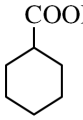
Product A is hexachlorobenzene

Product B is benzenehexachloride

Final Answer : Option (1)

90. **Ans (1)**

XI NCERT (Alkane)



cyclohexane carboxylic acid

91. **Ans (2)**

NCERT XI Page No. # 67

92. **Ans (2)**

NCERT XI Page No. # 67

93. **Ans (2)**

NCERT XI Page No. # 61

94. **Ans (4)**

NCERT XI Page No. # 59,60,62,63

95. **Ans (1)**

NCERT XI Page No. # 61

96. **Ans (2)**

NCERT XI Page No. # 62, 63

97. **Ans (2)**

NCERT XI Page No. # 59,60

98. **Ans (4)**

NCERT Pg # 64

99. **Ans (4)**

NCERT XI Page No. # 72

100. **Ans (2)**

NCERT Pg. # 76

101. **Ans (1)**

NCERT Pg. # 74

102. **Ans (3)**

NCERT XI Pg # 73

103. **Ans (3)**

NCERT XI, Pg # 73,74,75

104. **Ans (4)**

NCERT XII, Pg. # 9, 10

105. **Ans (2)**

NCERT XII Pg. No. # 13

106. **Ans (3)**

NCERT-XII, Pg. # 27,29,33,35,36

107. **Ans (4)**

NCERT Pg. # 6, 7, 9, 21

108. **Ans (3)**

NCERT, Pg. # 22

109. **Ans (1)**

NCERT Page No. 19

110. **Ans (2)**

NCERTXII Page No. 27

111. **Ans (3)**

NCERT XII Page No. # 15

112. **Ans (2)**

NCERT XII Page No. # 22,23

113. Ans (3)

Explaining the Question: This question asks which statement about chloroplast structure/function is incorrect.

Concept: Chloroplast – structure and function.

Solution: – Statement 3 is wrong because chloroplasts do show division of labour:

– Grana performs light reactions.

– Stroma performs dark reactions (CO_2 fixation).

There is division of labour in chloroplast, so this statement is incorrect.

Final Answer: Option (3)

114. Ans (2)
NCERT Pg. No. # 212

115. Ans (1)
NCERT, Pg. # 139, 140

116. Ans (1)
NCERT, Pg # 150

117. Ans (2)
NCERT-XI, Pg. # 209, 210, 211

118. Ans (3)
NCERT, Pg # 142-147

119. Ans (3)
NCERT, Pg.# 210

120. Ans (1)
NCERT, Pg. # 139

121. Ans (4)
NCERT XI Pg. # 210, 13.4

122. Ans (4)
NCERT pg no # 137, 147

123. Ans (1)
NCERT XI, Pg. # 234, 235

124. Ans (3)
NCERT, Pg. # 233

125. Ans (4)
NCERT XI Pg. # 159 [Fig. 12.3]

126. Ans (2)
NCERT Pg. # 161, 163, 164

127. Ans (3)
Explanation :NADH Production:3 NADH × 3 ATP / NADH = 9 ATP FADH2 Production:1 FADH2 × 2 ATP / FADH2 = 2 ATP GTP Production:1 GTP = 1 ATP

Total ATP per turn of the Krebs cycle:

9 ATP (from NADH) + 2 ATP (from FADH2) + 1 ATP (from GTP) = 12 ATP

Final answer : 3

128. Ans (4)
NCERT XI, Pg. # 231, 232

129. Ans (2)
NCERT XI, Pg. # 230

130. Ans (3)
NCERT XI Pg. No :- 241, 243, 244

131. Ans (2)
NCERT XI, Page No. 252, 253

132. Ans (1)
NCERT, Pg. # 247

133. Ans (1)
NCERT Pg. No. # 176, 177

134. Ans (2)
NCERT Pg. # 248,249

135. Ans (3)
NCERT Pg. # 168, 169

136. Ans (3)
NCERT Pg # 232, 233

137. Ans (4)
NCERT XI, Pg. # 235

138. Ans (3)
NCERT XI, Pg. # 236

139. Ans (2)
NCERT XI, Pg. # 233

140. Ans (2)
NCERT XI, Pg. # 236

141. Ans (2)
NCERT XI, Pg. # 236

142. Ans (2)
New NCERT Pg # 236

143. Ans (2)
NCERT Pg. No. # 241, 244, 245

144. Ans (2)
NCERT Pg. No. # 246

145. Ans (1)
NCERT Pg # 241

146. Ans (2)
NCERT Pg # 242, 243

147. Ans (2)
NCERT Pg # 242

148. Ans (2)
NCERT Pg # 243

149. Ans (3)
NCERT Pg # 244

150. Ans (3)
NCERT Pg # 248

151. Ans (2)
NCERT Pg. # 312

152. Ans (3)
NCERT Pg. # 308

153. Ans (2)
NCERT Page-309, 311

154. Ans (3)
NCERT, Pg. # 311

155. Ans (3)
NCERT, Pg. # 222-223

156. Ans (1)
NCERT XI Pg # 226

157. Ans (2)
NCERT XI Pg # 217-218

158. Ans (4)
NCERT Pg # 32, 33

159. Ans (2)
NCERT Pg. # 27

160. Ans (3)
NCERT Pg. # 26

161. Ans (1)
NCERT Pg. (E) # 26
NCERT Pg. (H) # 28

162. Ans (2)
NCERT Pg. # 27

163. Ans (2)
Module

164. Ans (3)
NCERT, Pg. # 44

165. Ans (2)
NCERT, Pg. # 48

166. Ans (4)
NCERT, Pg. # 38

167. Ans (3)
NCERT Pg. # 38

168. Ans (3)
Progesterone from corpus luteum maintains the endometrium for pregnancy.

169. Ans (3)
NCERT, Pg. # 34

170. Ans (3)
NCERT pg no # 33

171. Ans (1)
Module

172. Ans (2)
NCERT Pg. # 32, 33

173. Ans (4)
NCERT Pg # 37

174. Ans (1)
NCERT, Pg # 30

175. **Ans (4)**
NCERT Pg. # 33

176. **Ans (2)**
A, C and D

177. **Ans (2)**
Module

178. **Ans (2)**
NCERT, Pg. # 44-48

179. **Ans (3)**
NCERT Pg. # 44-45

180. **Ans (4)**
NCERT Pg. No. # 45